



EMIIA.AI
SPATIAL INTELLIGENCE

EMIIA.AI SIP – distributed cloud AI platform that solves key challenges in cloud technologies, IoT and RTLS systems, delivering **efficiency 10 times higher than global competitors.**

Result: Significant reduction of capital and operational expenses (**CapEx and OpEx**), as well as total cost of ownership (**TCO**) across various industries.

PROBLEM

- › Digital twins and AI, including AI agents, are expected to accelerate **internet traffic growth by 40% to 60%**
- › **By 2030, the volume of data is expected to reach 400 zettabytes** — 10,000 times more than in 2010. Internet traffic is doubling every 4 years
- › **Growth of capital, operating expenses, and total cost of ownership** (CapEx/OpEx/TCO) related to data processing, storage, delivery, as well as AI inference and software deployment.

SOLUTION

EMIIA.AI MAP (data mapping technology) — automation of working with digital twins of spatial objects and business processes. Processing, optimization, and compression of data while preserving the accuracy of the core information.

Up to 80% data volume is reduced, processing speeds up, and total cost of ownership (TCO) decreases.

Application:

- › Cloud technologies and AI
- › Internet of Things (IoT/AIoT)
- › RTLS system(d)

PROBLEM

› **The network is becoming more expensive than energy:** Equinix in Singapore pays \$1 million more for networks than for energy, while Cloudflare in Amsterdam pays \$2.1 million for networks versus \$1.5 million for electricity

› **Network latency** during data delivery, software deployment, and AI inference (east-west traffic, north-south)

› **AI autonomy**

› **Growth of capital, operational expenses, and total cost of ownership** (CapEx/OpEx/TCO) related to data processing, storage, and delivery, as well as to AI inference and software deployment

SOLUTION

EMIIA.AI LEM/IoT — distributed AI software-hardware infrastructure based on autonomous clusters and gateways EMIIA.AI LEM / EMIIA.AI IoT

Infrastructure, security, and communication channel costs are reduced by up to **60% (TCO – total cost of ownership)**

Two cycles of technological and economic equipment operation.

Application:

- › Cloud technologies and AI
- › Internet of Things (IoT/AIoT)
- › RTLS system

PROBLEM

- › **Lack of fully functional indoor positioning systems** in buildings and structures
- › **Searching for people during emergencies** in buildings and structures is difficult or even impossible
- › **Digital twins for RTLS systems are heavy**, complex to integrate, expensive to develop, and require ongoing maintenance costs
- › **Increase in capital, operational expenses, and total cost of ownership** (CapEx/OpEx/TCO) in the field of Internet of Things and RTLS systems

SOLUTION

EMIIA.AI SDK/MRV — seamless geolocation and navigation technology for open spaces and indoor environments (RTLS)

Sensor usage in Internet of Things and RTLS systems **is reduced by up to 30%**

EMIIA.AI MAP (data mapping technology) — automation of working with digital twins and their storage

Data volume is reduced by up to 80% and processing speeds up

Application:

- › Internet of Things (IoT/AIoT)
- › RTLS system

ECONOMIC EFFECTS

Reducing costs for storage, processing, deployment, and data delivery allows **businesses, researchers, and developers to:**

- › Significantly reduce TCO (total cost of ownership)
- › Increase ROI (return on investment)
- › Speed up time-to-market for products (Time-to-Market, T2M)

Business model - B2C, B2B, B2G, B2M:

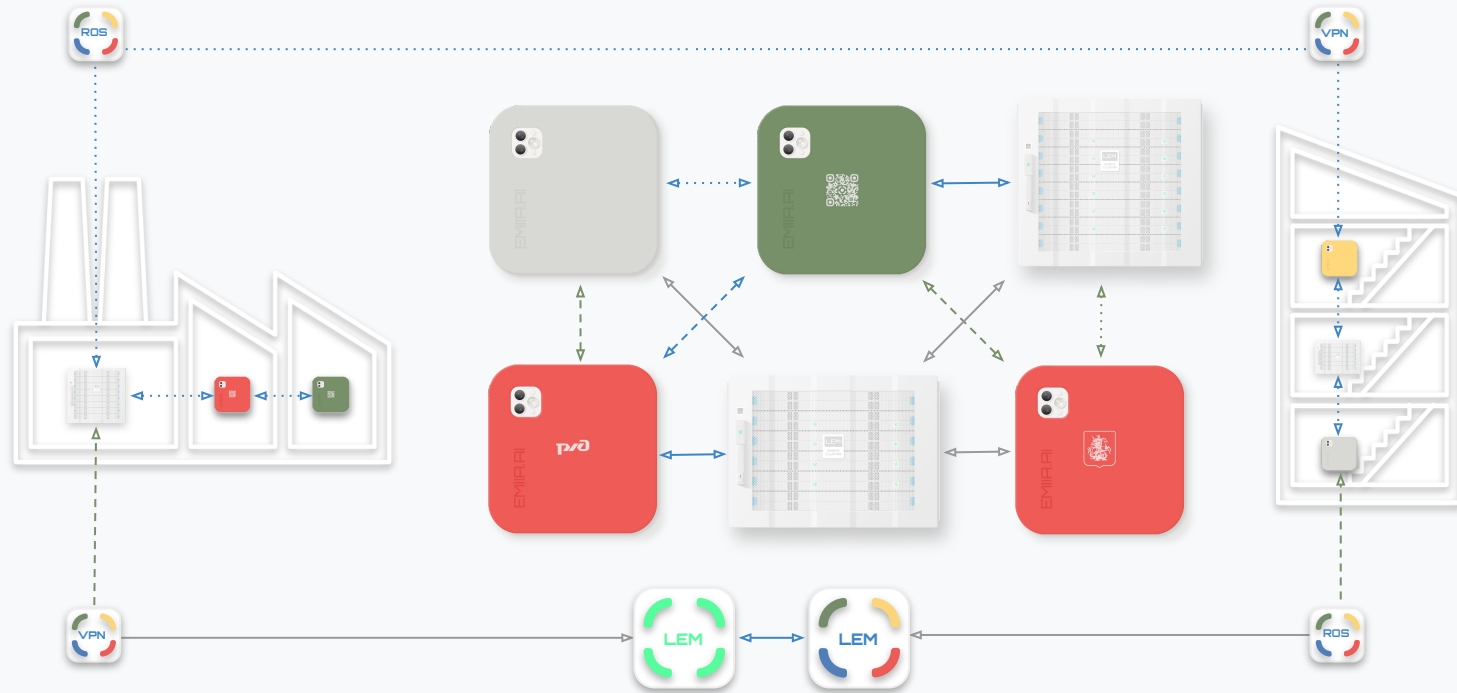
- › API as a product (APIaaS)
- › Artificial Intelligence as a Service (AIaaS)
- › Hardware as a Service (HaaS)

Market: BRICS+

Technology focus: Internet of Things (IoT/AIoT/QIoT)

Platform classification: Fault-tolerant autonomous infrastructure for AI, connectivity, and navigation (RTLS)

ARCHITECTURE OF THE DISTRIBUTED CLOUD AI PLATFORM EMIIA.AI SIP BASED ON AUTONOMOUS EMIIA.AI IoT GATEWAYS, CLUSTERS, AND EMIIA.AI LEM DATA CENTERS



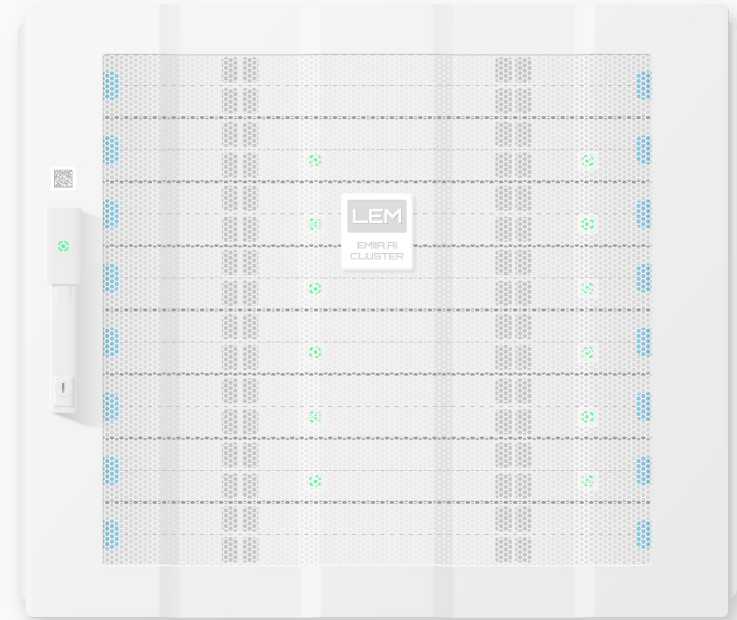
EMIIA.AI LEM CLUSTERS AND DATA CENTERS (HIBRID HPC/COLD, FROZEN DATA)

SOFTWARE AND HARDWARE SOLUTIONS OF THE DISTRIBUTED CLOUD AI
PLATFORM EMIIA.AI SIP: EMIIA.AI IoT GATEWAYS AND EMIIA.AI LEM CLUSTERS



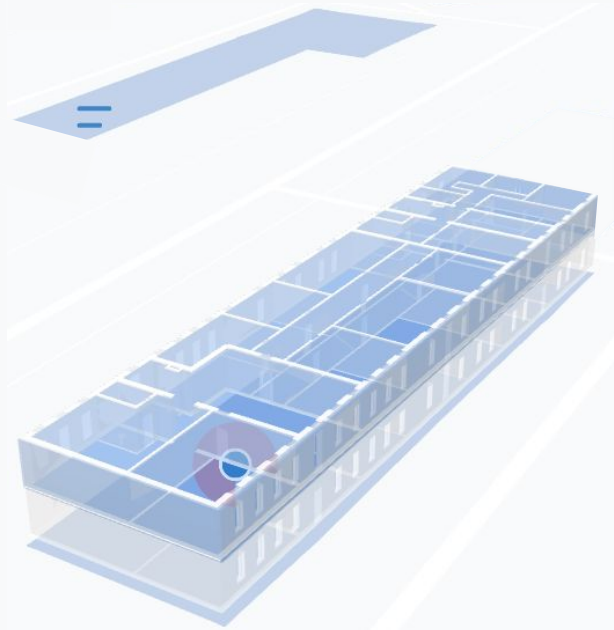
GATEWAY EMIIA.AI IoT

EDGE/FOG/CLOUD COMPUTING
◀────────────────────────────────▶
LAN/WAN/WLAN/WMN



CLUSTER EMIIA.AI LEM (NODE 12)

BASIC TECHNOLOGY



EMIIA.AI SIP: [SOS] ID: [R #3D4588] >

ID: [V #3D85C6] [37.17402, 55.97708] [2]



EMIIA.AI SDK/MRV — Machine Radio Vision
EMIIA.AI MRV (Machine Radio Vision), a neural network library for signal processing and data visualization (SDK/API).

FEATURES: Pattern recognition, calculation of speed, coordinates, and direction of moving objects, including behind radio-transparent obstacles (people, animals, etc.). Range: through radio-transparent barriers up to 9 meters, in open space up to 300 meters.

The solution enables precise real-time object tracking and intelligent AI-powered navigation — both outdoors and indoors.

EMIIA.AI MRV (RTLS system). The technology is based on radio-frequency machine vision using AI algorithms.

INTEGRATION: The technology forms the core of the IT architecture, hardware software stack, and EMIIA.AI SIP cloud platform.

USE CASES / PRODUCTS

AI AGENTS EMIIA.AI LLM

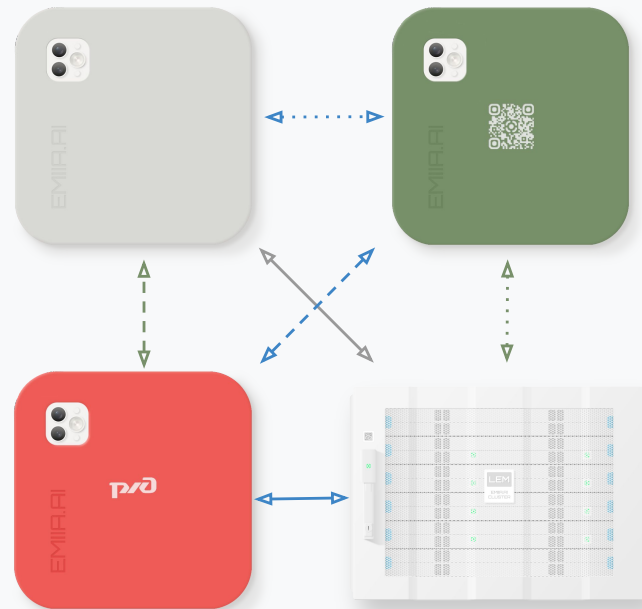
Autonomous AI agents within the ERP system structure: AI Mapper (digital twins), AI Auditor (production processes), AI Analyst (business processes), AI Pilot (coordination and creation of AI agents), AI Rescuer (public safety)...

NETWORK AI COMPUTER EMIIA.AI COM

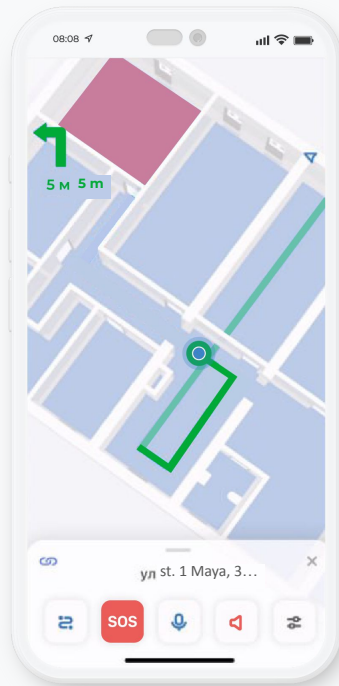
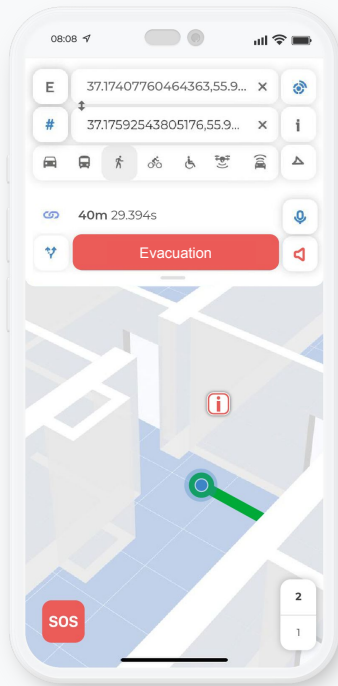
A powerful AI computer with wireless access to a cloud desktop (VDI) via thin clients, suitable for studying, research, gaming, programming, neural network training and inference, software deployment, as well as various professional and practical tasks.



[VIDEO PRESENTATION \(MP4\) »](#)



USE CASES / PRODUCTS



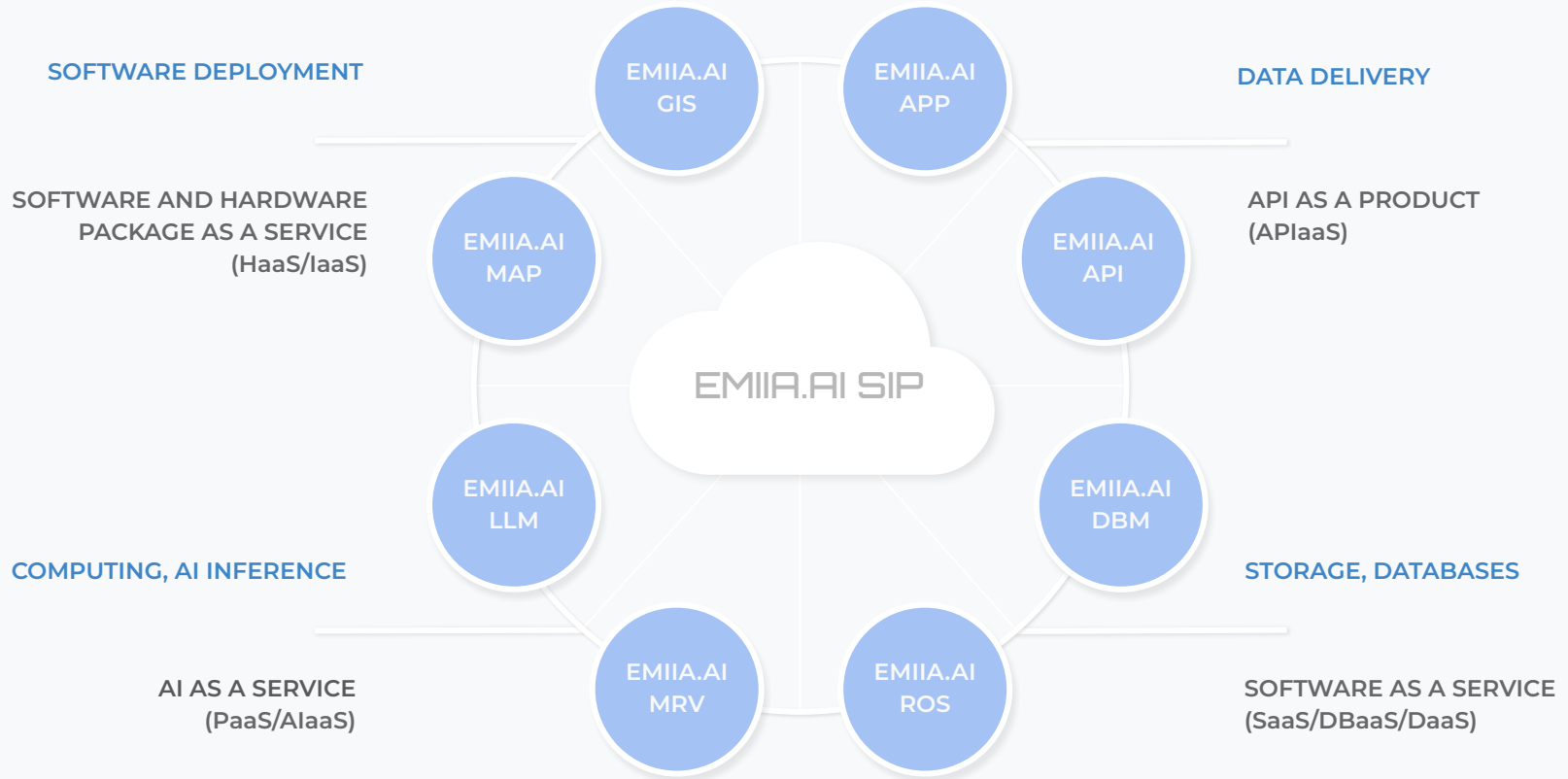
NAVIGATION WITHOUT GPS EMIIA.AI SDK/MRV

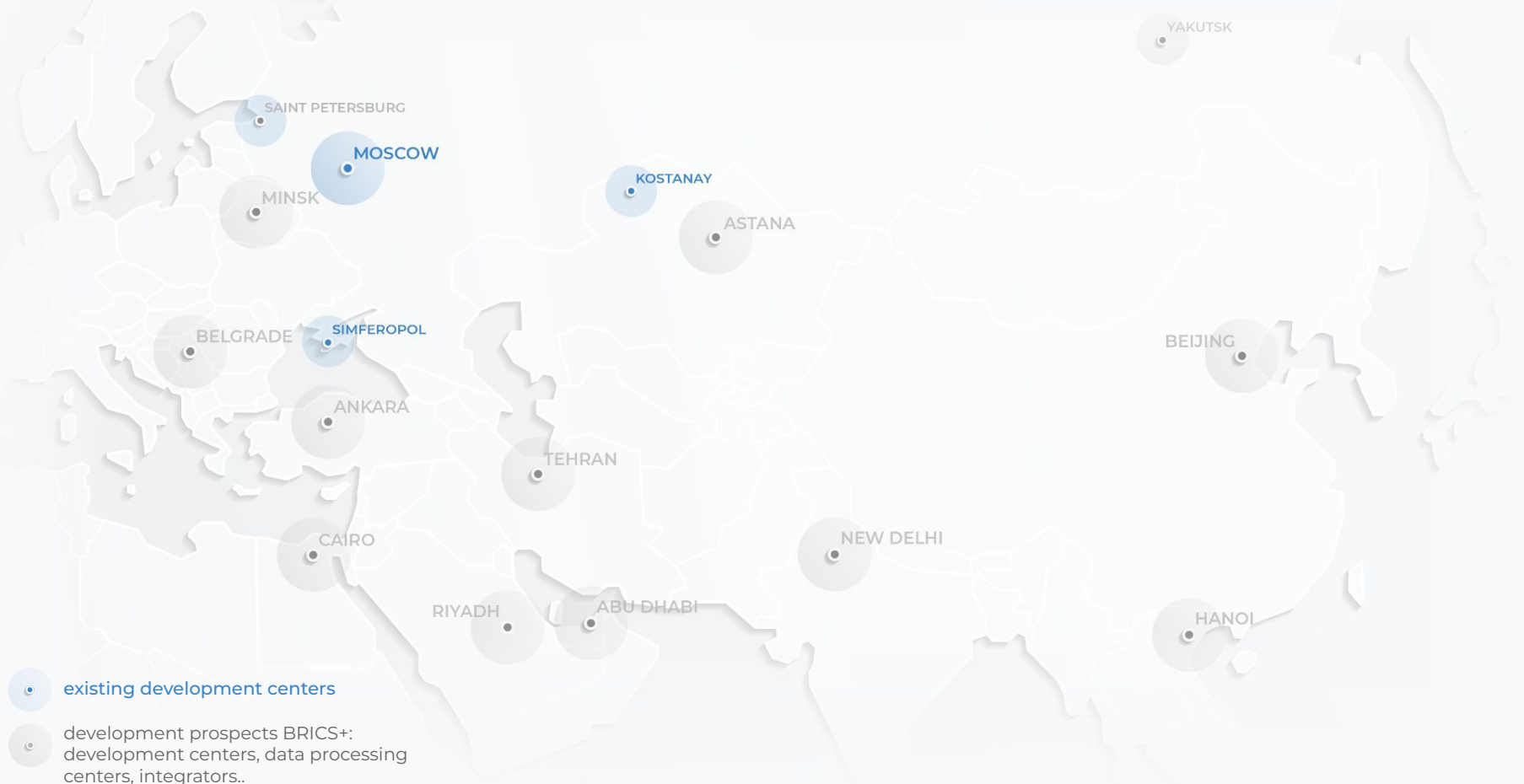
Indoor/Outdoor - navigation without using satellite systems — only access to our wireless network or the internet is needed to determine location, both outdoors and indoors.

SECURE ENVIRONMENT EMIIA.AI SOS EMIIA.AI

the first project that solves the problem of evacuation in emergencies using artificial intelligence.

SOFTWARE TOOLS OF THE DISTRIBUTED CLOUD AI PLATFORM EMIIA.AI SIP





- existing development centers
- development prospects BRICS+:
development centers, data processing
centers, integrators..

STRATEGIC PARTNERS



Lomonosov Cluster — a place where science turns into business. The cluster's main goal is to ensure close cooperation between research organizations, universities, and the real economy sector, as well as to create infrastructure for innovation.



RZD is Russia's largest railway network, transporting 1.1 billion passengers annually. "Transtelecom," a subsidiary of RZD, is in the top 5 broadband operators, managing 78,000 km of fiber optic cable and developing IT infrastructure.



Sber is Russia's leader in AI development: the bank invests about \$1 billion annually in artificial intelligence and plans to increase its investments by 1.5 times over the next 5 years in AI and IT infrastructure.

ABOUT US

"EMIIA LLC" is a DeepTech startup company specializing in end-to-end digital technologies. **TRL: 8-9, CAGR: 58%, R&DC: 57%.** Skolkovo IC Startup Registry: **ORN 1123966.** Registry of Small Technology Companies: **RST 1313.** Registry of Startups and Technology Companies of the City of Moscow: **STK 1233191.**

Strategy by 2030: to become one of the top 100 global and top 10 Russian companies in distributed spatial data processing systems, with a valuation exceeding \$1 billion. Achieve over thirty international and Russian patents.

We base our development not on analogs, but on technological trends.

Our mission is to create accessible AI infrastructure and provide unique resources for businesses, researchers, and developers in the field of operational technologies!

Promising directions for 2030+ include building quantum Internet of Things (QIoT) infrastructure — navigation based on quantum sensors (gravity, magnetic field), computing, communications, and data protection.

The EMIIA team of specialists has developed and commercially implemented the "KSK GRAAD": [GitHub ↗](#). Over one hundred thousand replications of the software solution (microprogram/firmware) have been installed. The commercial and scientific foundation formed in "KSK GRAAD" underlies the EMIIA project and technologies. EMIIA's source code is included in the program [GitHub Arctic World Archive ↗](#), aimed at archiving and preserving modern open-source software for a thousand years in permafrost conditions.

The project team, built around the core of research and development, consists of top-tier engineers who have worked together for many years in high-tech fields and is strengthened by AI agents.

[Our R&D team includes over 20 engineers.](#)

RATINGS OF INTERNATIONAL EXPERT RESOURCES

TOP 30

AI Companies and Startups in Russian Federation 2025 ↗

FINALIST/LONGLIST

BRICS Solutions Awards 2020/2024 ↗

TOP 10

Artificial Intelligence Companies in Russian Federation 2025 ↗

PROJECT TEAM (LEADING SPECIALISTS)



VLADIMIR STAROSTIN (CEO/CTO IT)

IT: ML, MRV, IT architecture, business model, commercialization

- Ten years of experience in IT management and development
- Two completed projects (software and hardware solutions)
- More than one hundred thousand replications of developed software (firmware)



ALEXANDRA SMYSLOVA (COO)

Industrial design: UI,UX, business model, commercialization

- Ten years in industrial design and project management
- More than one hundred interfaces and design solutions have been developed with more than one million replicated



ALEXEY LUMAN (CTO CLOUD)

Cloud: IT, ML, MRV, Cloud architecture

- Ten years of experience in IT management and development
- Two completed projects (software and hardware solutions)
- More than one hundred thousand replications of developed software (firmware)



ANDREY KONSTANTINOV (CDO)

Hardware solutions: ML, MRV, Hard architecture

- Ten years of experience in developing hardware solutions in the field of passive identification of moving objects.
- More than ten software and hardware systems have been developed

[ABOUT THE PROJECT AND TEAM \(INTERNET RESOURCE\) »](#)

EMIIA.AI

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